

# DEEKSHA GAUTAM

Software Engineer | AI Integrations, LLM Systems & Enterprise Applications

(+91) 9971077462 | gautamdeeksha7098@gmail.com | LinkedIn | GitHub | Portfolio

## SUMMARY

Software Engineer with **4 years of experience** building enterprise applications, including AI-integrated systems. Experienced in **LLM integration, RAG, model evaluation, hybrid AI architectures, privacy-aware routing, AI guardrails, agentic automation, backend engineering, CI/CD, and production deployment**. Strong foundation in Python, Java, APIs, Kafka, Spring Boot, and enterprise DevOps.

## TECHNICAL SKILLS

**AI & LLM:** LLM Integration, RAG, LLM Evaluation, Model Routing & Orchestration, AI Guardrails, Privacy-Aware AI, Prompt Engineering, Ollama, Gemini API, GroqCloud, Gemma 3, GPT-OSS

**AI Engineering:** Vector Databases (ChromaDB), Vector Search, Hybrid Retrieval, BM25, RRF, Embeddings, Document Ingestion, Metadata-Aware Chunking, Agentic Workflows

**Backend & Data:** Python, Java, SQL, FastAPI, Spring Boot, REST APIs, Kafka, MongoDB, PySpark, Data Pipelines

**DevOps & Engineering:** Docker, GitHub Actions, Jenkins, Harness, Bitbucket, Maven, SonarQube, OpenShift, CI/CD, Software Architecture, Security, Scalability, Production Deployment, Agile/SDLC

## PROJECTS

### CareAmy – AI Technical Advisor

Work in Progress | Production-Grade Local AI | Hybrid Retrieval

- Designing a **production-grade, fully local AI chatbot for EU pharmacies** with local inference and retrieval, restricting responses to pharmacy-configured answers.
- Developing a **hybrid retrieval pipeline** to match natural-language patient queries against configured Q&A, rank candidate matches, and return controlled responses.

### Local RAG Evaluation Lab

Python | ChromaDB | Ollama | BM25 | RRF

- Built a **local-only RAG pipeline** with document ingestion, metadata-aware chunking, local embeddings, **ChromaDB vector search, BM25 retrieval, and Reciprocal Rank Fusion**, generating grounded answers with citations.
- Evaluated **17 retrieval cases at Recall@4**; achieved **100% document hit rate, page hit rate, and Recall@4**, with hybrid **MRR of 0.893**.

### Hybrid AI Gateway

Python | FastAPI | Docker | Ollama | Groq | GitHub Actions

- Built a **policy-driven local-cloud AI gateway** with privacy classification, model routing, guardrails, output validation, policy-safe fallback, and operational metrics.
- Developed a **106-test evaluation suite** covering routing, privacy enforcement, fallback behavior, and live model execution.

### Local vs Cloud AI Lab

Python | Ollama | GroqCloud | LLM Evaluation

- Built a **34-request LLM benchmark** evaluating instruction following, structured output, grounded QA, reasoning, latency, and cost.
- GPT-OSS achieved **100% grounded QA and reasoning accuracy**, compared with **90% grounded QA and 80% reasoning accuracy** for local Gemma; cloud inference cost was approximately **\$0.00005/request**.

## PROFESSIONAL EXPERIENCE

### Technology Analyst II (C-10)

2023 – Present

Citi

- Architected an automated **three-level fund discount approval system** for NAM, integrating Citi applications with Salesforce and routing requests through approval matrices; reduced turnaround time by **78%**.
- Built an **AI agent for production change management** that creates change requests and coordinates production documentation across **Confluence, runbooks, and Tenant Ops**, reducing human intervention by **75%**.
- Developed a **Kafka-powered Direct Placement Investment platform**, contributing to **47% NAM client growth** and **30% higher activation**.
- Delivered an **AI-enabled document-processing workflow** through production release, achieving **92% customer satisfaction**; received the **Platinum Award**.

### Technology Analyst (C-09)

2022 – 2023

Citi

- Developed and maintained **35 TradeLink services** and independently migrated them to an upgraded **Spring Boot framework**, improving security, maintainability, and code efficiency with no production issues.
- Built an **AI-powered PR security plugin integrated with Devin.ai** that searches repositories for vulnerabilities when pull requests are raised and surfaces remediation options to developers.
- Migrated CI/CD from **Jenkins-RLM/Bitbucket** to **LSE-Harness/GitHub**, integrating **Maven, SonarQube, and OpenShift**.
- Independently deployed **35 applications across APAC, NAM, and EMEA**, owning release execution and validation; automated workflows reduced manual effort by **65%**; received the **Silver Award**.

## EDUCATION

MS in Analytics, Georgia Institute of Technology

2025 – 2027 (Expected)

B.Tech in Computer Science, Indira Gandhi Delhi Technical University for Women

2018 – 2022